Wk 2 Supervision Problems 2

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Set Operations

1. Let A a set. We define the *complement* of A, denoted A', to be the set of all elements $x \notin A$. Prove the identities

$$(A \cup B)' = A' \cap B',$$
$$(A \cap B)' = A' \cup B'.$$

2. Use the above to help you simplify the following expressions.

   (i) $((A \cup B) \cap (C \cup D))'$

   (ii) $A \cap (B \cap C')'$

   (iii) $(A \cap (B' \cup (C \cap D)'))'$

Analysis

1. Let $a, b \in \mathbb{R}$ be real numbers such that $|a|, |b| \geq k > 0$. Show that

$$\left|\frac{1}{a} - \frac{1}{b}\right| \leq \frac{|a-b|}{k^2}$$

2. Here is an answer to Q6 on the analysis exercises (proving the Greatest Lower Bound Property), but with some key details missing. Work through the proof yourself to fill in the gaps.

> First note that if [ ] for a set A, then [ ] for the set $-A$.
> Suppose S is [ ] that is bounded below. Let [ ] for S. Then $-S$ is [ ] which is [ ].
> By the [ ] axiom, $-S$ has a [ ], [ ].
> It follows that [ ] for S. Furthermore, because [ ], and since [ ] was arbitrary, we conclude that [ ] for S.